

**PETRONAS**

PETRONAS Tutela Coolant LL Premix

Designed to be the Fluid Technology Solutions of choice providing excellent protection and longer fluid life for Vehicle cooling systems

Description and Applications

PETRONAS Tutela Coolant LL Premix is a premium performance ready to use Long-Life Coolant delivering optimized performance and component durability for vehicles cooling systems. No further dilution is required. Formulated with Mono-Ethylene Glycol enhanced with a special Organic Acid salts inhibitor (OAT Technology), free of nitrites, borates, phosphates, silicates and amines, it provides excellent protection against overheating, frost and corrosion.

Benefits

PETRONAS Tutela Coolant LL Premix enhances your vehicle availability while significantly reducing your operating costs through:

- **Long Life Technology** to extend the fluid service life, preventing from deposit formation
- **Extended Cooling System Life** driven by:
 - o Excellent corrosion resistance to protect metal-based components
 - o Enhanced protection against freezing when outside temperature is down to -18°C
 - o Enhanced protection against boiling in case of circuit temperature up to 103°C
 - o Very good elastomer and plastic compatibility for a longer component's life

Provides outstanding Driving Experience through:

- **Excellent heat transfer properties to prevent overheating**
- **Extended Service life:**
 - Passenger vehicles and light commercial transporters 200,000km - 5 years
 - Heavy duty vehicles – 500,000km/6 years, /12,000 hrs.

Approvals, Specifications and Recommendations

Specifications:

• ASTM D3306 (and D4656)	• UNE 26-361-88/1	• Komatsu
• MAN 324 (Diesel Engines)	• Mack-RVI	• Hyundai
• ASTM D4985 (and D5345)	• Cummins 90 T8-4	• Subaru
• ASTM D6210 VOLVO	• Caterpillar EC-1	• ISUZU
• JIS K2234 (Japan) LLC	• E/L-1415b (MIL-Italy)	• SAAB
• JASO M325 (Japan) LLC	• NATO S-757	• Honda
• SAE J1034	• FVV Heft R443 (Germany)	• Toyota TS K2601 G
• Mercedes-Benz Sheet 325.3	• BT-PS-606A (MIL-Belgium)	• Mitsubishi ES-X64216
• Audi/SEAT/Skoda	• Detroit Diesel/Perkins	• Mazda MES 121C
• BMW N600 69.0	• General Motors GM 6277M	• Afnor R15-601
• Ford WSS-M97B44-D	• General Motors GM 1899M	• BR 637
• Volkswagen VW TL774D	• FSD 8704 (MIL-Sweden)	• Daihatsu
• Scania (Tech Info. TI 02-98)	• Navistar	• Önorm V5123
• BS 6580 (British Standard)	• DAF-Leyland 74001	• DCEA 615 (MIL-France)

Note: Always consult your owner's manual to check recommendations for your specific vehicle
It is recommended that Long Life Coolant Pre-mix not be mixed with other coolants in order to maintain performance claims.

Typical Physical Data

Parameters	Method	Unit	Typical Value
Color	-	-	Green
Relative Density @20°C	DIN 51757/4	g/cm ³	1.05
pH	ASTM D 1287	-	7.8
Boiling point	ASTM D 1120	°C	103
Freezing point	ASTM D 1177	°C	-18
Glassware corrosion test	ASTM D1384	-	PASS

All technical data are provided for reference only. These characteristics are typical of current production. Whilst future production will conform to PLI's specification, variations in these characteristics may occur.

Health, Safety and Environment

This product is unlikely to present any significant health and safety hazards when used in the recommended application. Avoid contact with skin. Wash immediately with soap and water after skin contact. Do not discharge into drains, soil or water.

For further detail regarding storage, safe handling, and disposal of product, please refers to product SDS or contact us at: www.pli-petronas.com

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